

Digital Signal Analysis and Processing Numerical Companion

CT 704 (BEI/BCT, IV-I) / EX 753 (BEX, IV-II) · TU, IOE · tiers over 45 papers, 2066–2082 BS

Ch 6 • IIR Filter Design

Numerical Problems & Solutions

47 of the 50 questions are one of three calculations and the other 3 are the spectral transformation, which is also below. Every design paper in the 45 is here: 32 bilinear Butterworth, 4 bilinear Chebyshev, 11 impulse invariance (two of them sharing a statement with a bilinear paper), 3 spectral transformation.

The seven steps, used by every problem in §1, §2 and §3

1. Get ω_p, ω_s in rad/sample. If the paper gives Hz, use $\omega = 2\pi f / f_s$. If it gives them in π already, do nothing.
2. Get α_p, α_s in positive dB. From a gain, $\alpha = -20 \log_{10} |H|$. From a ripple δ_p , the edge gain is $1 - \delta_p$ first. A deviation already in dB is used as printed.
3. Translate to the analog domain, by the rule of the method: **bilinear** $\Omega = \frac{2}{T} \tan \frac{\omega}{2}$ (pre-warp), **impulse invariance** $\Omega = \omega / T$.
4. Order $N = \left\lceil \frac{\log [(10^{0.1\alpha_s} - 1) / (10^{0.1\alpha_p} - 1)]}{2 \log(\Omega_s / \Omega_p)} \right\rceil$, and cut-off $\Omega_c = \Omega_p / (10^{0.1\alpha_p} - 1)^{1/2N}$.
5. Poles $s_k = \Omega_c e^{j\pi(N+2k+1)/2N}$, $k = 0 \dots N - 1$, then $H_a(s) = \Omega_c^N / \prod_k (s - s_k)$, written as real quadratic sections.
6. Map back: substitute $s = \frac{2}{T} \frac{1-z^{-1}}{1+z^{-1}}$, or expand in partial fractions and use $\frac{1}{s-p_k} \rightarrow \frac{1}{1-e^{p_k T} z^{-1}}$.
7. Check $|H(e^{j\omega_p})|$ and $|H(e^{j\omega_s})|$ against the specification. Two substitutions, and they catch almost every slip.

Checks that cost nothing

1. $H(z)$ from a bilinear design has all its zeros at $z = -1$, so its numerator is $K(1 + z^{-1})^N$ and its coefficients are K times the binomials $1, N, \binom{N}{2}, \dots$. **Anything else is an algebra error.**
2. $|H(e^{j0})| = 1$ exactly for a Butterworth design, either method. Set $z = 1$, i.e. add up the coefficients: $\sum b_k / \sum a_k = 1$.
3. $|H(e^{j\omega_p})|$ must come out at exactly $10^{-\alpha_p/20}$, because Ω_c was chosen to make the passband exact.
4. Every pole must satisfy $|z| < 1$, and they come in conjugate pairs, so all the printed coefficients are real.

The three shortcuts worth taking in the exam

1. **Set $T = 1$ in any bilinear design and say why.** The $2/T$ from the pre-warp and the $2/T$ from the substitution cancel, so $H(z)$ does not depend on T (§6.5 of the theory). Nine papers give a sampling frequency that changes nothing.
2. **Do not expand $\prod (s - s_k)$ into a single polynomial** unless asked. Leave $H_a(s)$ as a product of quadratics; the bilinear substitution then runs one quadratic at a time and the arithmetic stays small.
3. For $N = 2$ use the tabulated normalised form $H_a(s) = \frac{1}{s^2 + \sqrt{2}s + 1}$ with $s \rightarrow s/\Omega_c$; for $N = 3$, $\frac{1}{(s+1)(s^2+s+1)}$. **Both are just the pole formula pre-computed.**

1 Bilinear transformation, Butterworth prototype

TOP 32/45 80 Bh 12, 81 Bh 12+3, 81 Ba 2+10, 75 Ch, 70 Ch 10, 82 Ba 12+3, 78 Ch 10, 73 Ch 11+4, 73 Shr 12, 80 Ba 11, 79 Ba 11+3, 79 Ch 15, 72 Ash 12, 73 Bh 11, 71 Bh 9, 72 Ka 15, 71 Shr 15, 77 Ch 10, 73 Ma 11, 79 Bh, 71 Ch, 69 Ch 11+4, 76 Ash 12, 74 Ash 15, 66 Ma 15, 68 Bh 18, 75 Ash 12, 74 Bh 9, 76 Bh 1+10, 76 Ch 10, 70 Bh 9, 82 Bh 3+12 · **405 marks, the single biggest question in the subject**

QUESTION AS PRINTED 2079 Bhadra · 2071 Chaitra · 2069 Chaitra · [11+4]

Design a digital Butterworth low pass filter using the bilinear transformation method for $W_p = 0.25\pi$ rad, $W_s = 0.55\pi$ rad, $\delta_p = 0.11$, $\delta_s = 0.21$, sampling frequency 0.5 Hz. Then convert the obtained digital low-pass filter to a high-pass filter with new pass band frequency $W'_p = 0.45\pi$ using the digital domain transformation.

Step 1: the specification in dB

- $\delta_p = 0.11$ is a **deviation**, so the passband edge gain is $1 - 0.11 = 0.89$. $\delta_s = 0.21$ is already a **gain**.

$$\alpha_p = -20 \log_{10} 0.89 = \mathbf{1.0122 \text{ dB}}, \quad \alpha_s = -20 \log_{10} 0.21 = \mathbf{13.5556 \text{ dB}}$$

- Sampling frequency 0.5 Hz means $T = 2$ s. **It will cancel**, so take $T = 1$ and note it. (Working it through with $T = 2$ gives the identical $H(z)$; only the intermediate Ω values change, by a factor of 2.)

Step 2: pre-warp the band edges

$$\Omega_p = 2 \tan \frac{0.25\pi}{2} = 2 \tan(22.5^\circ) = \mathbf{0.828427}, \quad \Omega_s = 2 \tan \frac{0.55\pi}{2} = 2 \tan(49.5^\circ) = \mathbf{2.341699}$$

- **This is the step that is skipped**, and skipping it puts the finished band edges in the wrong place.

Step 3: order and cut-off

$$10^{0.1\alpha_p} - 1 = 10^{0.10122} - 1 = 0.262467, \quad 10^{0.1\alpha_s} - 1 = 10^{1.35556} - 1 = 21.675737$$

$$N \geq \frac{\log_{10}(21.675737/0.262467)}{2 \log_{10}(2.341699/0.828427)} = \frac{\log_{10} 82.5847}{2 \log_{10} 2.82668} = \frac{1.91690}{2(0.45128)} = 2.1239$$

$$\boxed{N = 3} \quad (\text{round up, always})$$

$$\Omega_c = \frac{\Omega_p}{(10^{0.1\alpha_p} - 1)^{1/2N}} = \frac{0.828427}{(0.262467)^{1/6}} = \frac{0.828427}{0.800164} = \mathbf{1.035321}$$

Step 4: the poles, and $H_a(s)$

- $N = 3$, so three left-half-plane poles at radius Ω_c , spaced $\pi/3$, at angles $\pi \frac{3+2k+1}{6}$ for $k = 0, 1, 2$, i.e. $120^\circ, 180^\circ, 240^\circ$:

k	Angle	$s_k = \Omega_c e^{j\theta}$	Factor
0	120°	$-0.517661 + j0.896615$	$s^2 + 1.035321 s + 1.071891$
2	240°	$-0.517661 - j0.896615$	
1	180°	-1.035321	$s + 1.035321$

- The conjugate pair multiplies to a real quadratic: sum of roots -1.035321 , and product $= \Omega_c^2 = 1.071891$. **For a Butterworth filter every quadratic section has constant term Ω_c^2** , which is a free check.

$$H_a(s) = \frac{\Omega_c^3}{(s + 1.035321)(s^2 + 1.035321 s + 1.071891)} = \frac{1.109751}{(s + 1.035321)(s^2 + 1.035321 s + 1.071891)}$$

Step 5: the bilinear substitution

- Put $s = 2 \frac{1 - z^{-1}}{1 + z^{-1}}$ into each factor, multiplying numerator and denominator of the whole expression by $(1 + z^{-1})^3$:

$$s + 1.035321 \rightarrow \frac{2(1 - z^{-1}) + 1.035321(1 + z^{-1})}{1 + z^{-1}} = \frac{3.035321 - 0.964679z^{-1}}{1 + z^{-1}}$$

$$\begin{aligned} s^2 + 1.035321s + 1.071891 &\rightarrow \frac{4(1 - z^{-1})^2 + 2.070642(1 - z^{-2}) + 1.071891(1 + z^{-1})^2}{(1 + z^{-1})^2} \\ &= \frac{7.142534 - 5.856219z^{-1} + 3.001248z^{-2}}{(1 + z^{-1})^2} \end{aligned}$$

- Multiply the two denominators:

$$21.679886 - 24.665755z^{-1} + 14.759120z^{-2} - 2.895239z^{-3}$$

- Divide through by the constant term 21.679886, and the $(1 + z^{-1})^3$ left over on top carries Ω_c^3 , giving $K = 1.109751/21.679886 = 0.051188$:

$$\begin{aligned} H(z) &= \frac{0.051188 (1 + z^{-1})^3}{1 - 1.137725z^{-1} + 0.680775z^{-2} - 0.133545z^{-3}} \\ &= \frac{0.051188 + 0.153564z^{-1} + 0.153564z^{-2} + 0.051188z^{-3}}{1 - 1.137725z^{-1} + 0.680775z^{-2} - 0.133545z^{-3}} \end{aligned}$$

- **The numerator coefficients are $0.051188 \times (1, 3, 3, 1)$** . If they are not the binomials, the substitution went wrong.

Step 6: check, before writing anything else

- At $z = 1$ every z^{-1} becomes 1, so $H(1)$ is just the sum of the numerator coefficients over the sum of the denominator's:

$$H(1) = \frac{0.051188(1 + 3 + 3 + 1)}{1 - 1.137725 + 0.680775 - 0.133545} = \frac{0.409504}{0.409505} = \mathbf{1.0000}$$

✓ **Two additions, and they catch a wrong K instantly.**

- $|H(e^{j0.25\pi})| = \mathbf{0.8900}$, which is $1 - \delta_p$ **exactly**, as the choice of Ω_c guaranteed. ✓
- $|H(e^{j0.55\pi})| = \mathbf{0.0861} \leq 0.21$. ✓ **Comfortably over-satisfied**, which is what rounding N up from 2.12 to 3 bought.

Step 7: the low pass to high pass conversion (the 4 marks)

- Prototype edge $\omega_p = 0.25\pi$, new edge $\omega'_p = 0.45\pi$:

$$\alpha = -\frac{\cos \frac{\omega_p + \omega'_p}{2}}{\cos \frac{\omega_p - \omega'_p}{2}} = -\frac{\cos(0.35\pi)}{\cos(0.10\pi)} = -\frac{0.45399}{0.95106} = \mathbf{-0.47735}$$

- Substitute $z^{-1} \rightarrow -\frac{z^{-1} + \alpha}{1 + \alpha z^{-1}} = -\frac{z^{-1} - 0.47735}{1 - 0.47735z^{-1}}$ into $H(z)$ above:

$$H_{hp}(z) = \frac{0.276238 (1 - z^{-1})^3}{1 - 0.683740z^{-1} + 0.457627z^{-2} - 0.068533z^{-3}}$$

$$= \frac{0.276238 - 0.828713z^{-1} + 0.828713z^{-2} - 0.276238z^{-3}}{1 - 0.683740z^{-1} + 0.457627z^{-2} - 0.068533z^{-3}}$$

- Checks: $|H_{hp}(e^{j0})| = \mathbf{0}$ exactly, because every zero moved from $z = -1$ to $z = +1$; and $|H_{hp}(e^{j0.45\pi})| = \mathbf{0.8900}$, the same $1 - \delta_p$ the prototype had at its own edge. **Both checks are free and both are worth writing down.**

Every other bilinear Butterworth paper

Same seven steps, different numbers. **T is taken as 1 throughout**, since it cancels, so a paper's own sampling frequency changes the Ω_c column but not the $H(z)$ column. Ω_p, Ω_s are the *pre-warped* edges. Every numerator is $K(1 + z^{-1})^N$ with $K = (\sum a_k)/2^N$.

Papers	Spec, as printed	α_p	α_s	N	Ω_c	$H(z)$ denominator
80 Bh, 81 Bh	0.9 to 0.5π ; 0.2 from 0.75π	0.9151	13.9794	3	2.54674	$1 + 0.43938z^{-1}$ $+ 0.38450z^{-2}$ $+ 0.041621z^{-3}$
81 Ba	0.6 to 0.35π ; 0.1 from 0.7π	4.4370	20	2	1.06140	$1 - 0.70699z^{-1}$ $+ 0.26136z^{-2}$
75 Ch, 70 Ch, 82 Ba	0.8 to 0.2π ; 0.2 from 0.6π	1.9382	13.9794	2	0.75037	$1 - 1.02819z^{-1}$ $+ 0.36508z^{-2}$
78 Ch	0.82 to 0.22π ; 0.18 from 0.58π	1.7237	14.8945	2	0.86185	$1 - 0.90725z^{-1}$ $+ 0.32103z^{-2}$
73 Shr, 77 Ch, 76 Bh	0.89125 to 0.2π ; 0.17783 from 0.3π (= 1 dB and 15 dB)	1	15	6	0.72729	see the cascade in §1.2

Papers	Spec, as printed	α_p	α_s	N	Ω_c	$H(z)$ denominator
80 Ba	0.26 π , 0.99 dB; −14.99 dB at 0.58 π	0.99	14.99	3	1.08611	$1 - 1.06340z^{-1} + 0.63628z^{-2} - 0.12114z^{-3}$
79 Ba	0.24 π , 0.98 dB; −14.95 dB at 0.57 π	0.98	14.95	3	0.99560	$1 - 1.19710z^{-1} + 0.71861z^{-2} - 0.14406z^{-3}$
79 Ch, 72 Ash	0.24 π , 1 dB; −14.9 dB at 0.57 π	1	14.9	3	0.99186	$1 - 1.20280z^{-1} + 0.72232z^{-2} - 0.14509z^{-3}$
73 Bh	0.25 π , 0.99 dB; −14.85 dB at 0.59 π	0.99	14.85	3	1.03961	$1 - 1.13140z^{-1} + 0.67685z^{-2} - 0.13245z^{-3}$
71 Bh	0.25 π , 1 dB; −15 dB at 0.55 π	1	15	3	1.03767	$1 - 1.13430z^{-1} + 0.67862z^{-2} - 0.13295z^{-3}$
72 Ka	0.25 π , 1 dB; −15 dB at 0.45 π	1	15	4	0.98086	$1 - 1.64760z^{-1} + 1.35630z^{-2} - 0.52524z^{-3} + 0.083384z^{-4}$
71 Shr	0.2 π , 1 dB; −15 dB at 0.4 π	1	15	3	0.81397	$1 - 1.48260z^{-1} + 0.92964z^{-2} - 0.20333z^{-3}$
73 Ma	0.2 π , 0.98 dB; 20 dB at 0.5 π	0.98	20	3	0.81704	$1 - 1.47760z^{-1} + 0.92551z^{-2} - 0.20214z^{-3}$
79 Bh, 71 Ch, 69 Ch	worked in full above	1.0122	13.5556	3	1.03532	$1 - 1.13773z^{-1} + 0.68078z^{-2} - 0.13355z^{-3}$

Papers	Spec, as printed	α_p	α_s	N	Ω_c	$H(z)$ denominator
76 Ash	0.22 π , 0.54 π ; $\delta_p=0.11$, $\delta_s=0.22$	1.0122	13.1515	2	1.00598	$1 - 0.76056z^{-1} + 0.27575z^{-2}$
74 Ash	0.27 π , 0.58 π ; $\delta_p=0.11$, $\delta_s=0.21$	1.0122	13.5556	3	1.12856	$1 - 1.00260z^{-1} + 0.60224z^{-2} - 0.11157z^{-3}$
68 Bh	0.25 π , 0.45 π ; $\delta_p=0.17$, $\delta_s=0.27$	1.6184	11.3727	3	0.94579	$1 - 1.27310z^{-1} + 0.77005z^{-2} - 0.15836z^{-3}$
75 Ash, 74 Bh	0.7 dB to 0.15 π ; 14 dB from 0.6 π	0.7	14	2	0.74249	$1 - 1.03700z^{-1} + 0.36853z^{-2}$

Every numerator is $K(1 + z^{-1})^N$ with K fixed by $|H(e^{j0})| = 1$, i.e. $K = (\sum a_k)/2^N$. The four papers whose calculation genuinely differs are split out below.

1.1 Three papers whose specification is given in Hz

HOT 3/45 76 Ch **10**, 82 Bh **3+12**, 70 Bh **9** · only step 1 changes

QUESTION AS PRINTED 2076 Chaitra · [10]

Design a digital Butterworth low pass filter using the bilinear transformation method for passband peak-to-peak ripple ≤ 1 dB and passband edge frequency 1.2 kHz; stopband attenuation ≥ 40 dB and stopband edge frequency 2.5 kHz; sample rate 8 kHz.

$$\omega_p = \frac{2\pi(1200)}{8000} = \mathbf{0.3\pi}, \quad \omega_s = \frac{2\pi(2500)}{8000} = \mathbf{0.625\pi}$$

- Pre-warp: $\Omega_p = 2 \tan(0.15\pi) = 1.01905$, $\Omega_s = 2 \tan(0.3125\pi) = 2.99321$.
- $\alpha_p = 1$, $\alpha_s = 40$. $N = 4.9010 \Rightarrow \boxed{N = 5}$, $\Omega_c = 1.16648$.
- Sections of $H_a(s)$: $(s + 1.16648)$, $(s^2 + 0.72093s + 1.36068)$, $(s^2 + 1.88741s + 1.36068)$, constant term $\Omega_c^2 = 1.36068$ in both quadratics.

$$H(z) = \frac{0.010975 (1 + z^{-1})^5}{(1 - 0.26323z^{-1})(1 - 0.77598z^{-1} + 0.57608z^{-2})(1 - 0.57782z^{-1} + 0.17359z^{-2})}$$

- Check: $|H(e^{j0.3\pi})| = 0.89125$ (exactly −1 dB) and $|H(e^{j0.625\pi})| = 0.00899 \leq 0.01$. ✓

82 Bh 3+12

350 Hz and 1000 Hz, -3 dB and -10 dB, $f_s = 5000$ Hz, plus "explain the frequency warping effect in detail" (§6.4 of the theory).

- $\omega_p = 2\pi(350)/5000 = 0.14\pi$, $\omega_s = 2\pi(1000)/5000 = 0.4\pi$; $\Omega_p = 0.44705$, $\Omega_s = 1.45309$; $\alpha_p = 3$, $\alpha_s = 10$.
- $N = 0.9340 \Rightarrow \boxed{N = 1}$, $\Omega_c = 0.44812$. **The smallest filter in the chapter:** one real pole, no quadratic section at all.

$$H_a(s) = \frac{0.44812}{s + 0.44812} \Rightarrow H(z) = \frac{0.183045(1 + z^{-1})}{1 - 0.633910z^{-1}}$$

- Check $|H(e^{j0.14\pi})| = 0.70795 \approx 10^{-3/20} = 0.70795$. ✓ With $\alpha_p = 3$ dB the passband edge is the 3 dB point, so $\Omega_c \approx \Omega_p$ and the order formula barely clears 1.

70 Bh 9

defective as printed

QUESTION AS PRINTED 2070 Bhadra · [9]

Design a digital low pass Butterworth filter by applying bilinear transformation technique for: pass band edge = 120 Hz, pass band attenuation = 1 dB, stop band edge = 170 Hz, stop band attenuation = 16 dB. Assume sampling frequency of 256 Hz.

- With $f_s = 256$ Hz the **Nyquist frequency is 128 Hz**. The stopband edge of 170 Hz is **above it**, so it does not exist as a digital frequency: $\omega_s = 2\pi(170)/256 = 1.3281\pi > \pi$.
- Pre-warping it gives $\tan(0.664\pi) < 0$, a **negative** Ω_s , and the order formula takes the logarithm of a negative number. **No filter meets this specification, because the specification is not a specification.**
- Aliased, 170 Hz lands at $256 - 170 = 86$ Hz, which is *below* the 120 Hz passband edge. The stopband would sit inside the passband.
- **Say that, then solve the workable reading:** treat the two edges as *analog* frequencies, skip the pre-warp, and map with $T = 1/256$.

$$\Omega_p = 2\pi(120) = 753.98, \quad \Omega_s = 2\pi(170) = 1068.14, \quad N = 7.19 \Rightarrow \boxed{N = 8}, \quad \Omega_c = 820.42$$

- An 8th-order filter for a 9-mark question is itself a sign the paper is wrong. **The marks are in naming the defect**, not in grinding out eight poles.

1.2 The textbook filter: 73 Shr, 77 Ch and 76 Bh

HOT 3/45 73 Shr 12, 77 Ch 10, 76 Bh 1+10 · one spec stated three ways

- 73 Shr prints gains: 0.89125 to 0.2π , 0.17783 from 0.3π . 77 Ch and 76 Bh print the same thing in dB: 1 dB and 15 dB. **They are the same filter**, and it is Oppenheim's own worked example.
- $\Omega_p = 0.64984$, $\Omega_s = 1.01905$, $N = 5.3044 \Rightarrow \boxed{N = 6}$, $\Omega_c = 0.72729$.
- $H_a(s)$, three quadratics, each with constant term $\Omega_c^2 = 0.52895$:

$$H(z) = \frac{0.147944}{(s^2 + 0.37647s + 0.52895)(s^2 + 1.02854s + 0.52895)(s^2 + 1.40502s + 0.52895)} \cdot \frac{0.00057969(1 + z^{-1})^6}{(1 - 1.31432z^{-1} + 0.71490z^{-2})(1 - 1.05406z^{-1} + 0.37532z^{-2})(1 - 0.94592z^{-1} + 0.23422z^{-2})}$$

- Check: $|H(e^{j0.2\pi})| = 0.89125$ exactly, $|H(e^{j0.3\pi})| = 0.13101 \leq 0.17783$. ✓
- **A textbook difference worth knowing.** Oppenheim places the 3 dB point so the *stopband* is met exactly, giving $\Omega_c = 0.76627$ and a different (equally valid) $H(z)$. The local formula meets the *passband* exactly, as above. **State which convention you used** and either is right.
- 76 Bh's extra mark is the Butterworth/Chebyshev comparison: on this very spec Chebyshev needs $N = 4$ against Butterworth's $N = 6$.

1.3 Two papers where the arithmetic collapses

PIN 2/45 73 Ch 11+4, 66 Ma 15

73 Ch 11+4

QUESTION AS PRINTED 2073 Chaitra · [11+4]

Design a digital Butterworth low pass filter using bilinear transformation for $0.707 \leq |H(e^{jw})| \leq 1$ for $0 \leq w \leq \pi/2$; $|H(e^{jw})| \leq 0.2$ for $3\pi/4 \leq w \leq \pi$; $T = 1$ s. Realize the filter using the most convenient realization form.

- 0.707 is $1/\sqrt{2}$, so the passband edge is the **3 dB point** and $\Omega_c = \Omega_p$ with no formula at all:

$$\Omega_p = 2 \tan \frac{\pi}{4} = 2, \quad \Omega_s = 2 \tan \frac{3\pi}{8} = 4.82843, \quad \Omega_c = \Omega_p = 2$$

- $\alpha_p = 3.0103$ dB, $\alpha_s = 13.9794$ dB, $N = 1.8026 \Rightarrow \boxed{N = 2}$.

$$H_a(s) = \frac{\Omega_c^2}{s^2 + \sqrt{2}\Omega_c s + \Omega_c^2} = \frac{4}{s^2 + 2\sqrt{2}s + 4}$$

- Substituting $s = 2\frac{1-z^{-1}}{1+z^{-1}}$ the whole thing stays **exact**:

$$H(z) = \frac{4(1+z^{-1})^2}{(8+4\sqrt{2}) + (8-4\sqrt{2})z^{-2}} = \frac{\frac{1}{2+\sqrt{2}}(1+z^{-1})^2}{1 + \frac{2-\sqrt{2}}{2+\sqrt{2}}z^{-2}} = \frac{0.292893(1+z^{-1})^2}{1+0.171573z^{-2}}$$

- **The z^{-1} term vanishes identically.** That happens only because $\omega_p = \pi/2$ puts the poles at $\pm 135^\circ$, symmetric about the imaginary axis, so their sum is purely imaginary and cancels.
- Taking 0.707 literally instead of $1/\sqrt{2}$ gives $\Omega_c = 1.99970$ and a residual $-0.000177z^{-1}$. **That is rounding, not a term.** Write the exact form.
- **The 4 marks for realization:** with one quadratic section, direct form II is the most convenient, and it needs 2 delays, 3 multipliers (the numerator's 1, 2, 1 are shifts and adds) and 4 adders. Cross-reference chapter 4.

66 Ma 15

$\omega_p = 0.3\pi$, $\omega_s = 0.4\pi$, $\delta_p = 0.11$, $\delta_s = 0.21$, and the paper asks for all five parts by name.

- (a) $\Omega_p = 1.01905$, $\Omega_s = 1.45309$, $\alpha_p = 1.0122$, $\alpha_s = 13.5556$; $N = 6.2199 \Rightarrow \boxed{N = 7}$. **The narrowest transition in the chapter** (0.3π to 0.4π) buys the highest order.
- (b) $\Omega_c = 1.01905/(0.262467)^{1/14} = \mathbf{1.12122}$.
- (c) The paper asks for the poles of the squared magnitude response, which is the full set of $2N = 14$: $s_k = \Omega_c e^{j\pi(2k+1)/14}$ for $k = 0 \dots 13$, equally spaced by $\pi/7 = 25.71^\circ$ on the circle of radius 1.12122. **Half are discarded**; only the 7 with $\sigma < 0$ belong to $H_a(s)$.
- (d) $H_a(s) = \frac{\Omega_c^7}{(s + 1.12122) \prod_{i=1}^3 (s^2 + b_i s + 1.25713)}$ with $b_i = 0.49899, 1.39814, 2.02037$ and $\Omega_c^2 = 1.25713$ throughout.
- (e) $H(z) = \frac{0.0015237(1+z^{-1})^7}{(1-0.28155z^{-1})(1-0.87700z^{-1}+0.68091z^{-2})(1-0.68117z^{-1}+0.30557z^{-2})(1-0.59000z^{-1}+0.13083z^{-2})}$
- Check $|H(e^{j0.3\pi})| = 0.8900$ and $|H(e^{j0.4\pi})| = 0.16074 \leq 0.21$. ✓

2 Impulse invariance

TOP 11/45 78 Bh 12+3, 72 Ch 3+12, 81 Ch, 80 Ch, 69 Bh 12, 70 Asa 13, 74 Ch 12, 67 Mng 15, 69 Bh 15, 73 Shr 3, 70 Ma 3+6 · 133 marks

Steps 1, 4 and 5 are unchanged from §1. Step 3 becomes $\Omega = \omega/T$ with no pre-warp, and step 6 becomes a partial fraction expansion. **T no longer cancels**, so it must be carried.

The pair that answers "compare the two methods" 12

75 Ash and 74 Bh print exactly the same specification as 74 Ch, and ask for the other method. Nothing else in the 45 papers demonstrates the difference so cleanly, so solve them together.

QUESTION AS PRINTED 2075 Ashwin / 2074 Bhadra (bilinear) · 2074 Chaitra (impulse invariance) · [12] [9] [12]

Design a digital Butterworth low pass filter whose passband magnitude is constant to 0.7 dB below the frequency 0.15π and whose stopband attenuation is at least 14 dB for frequencies between 0.6π and π .

Step	75 Ash / 74 Bh: bilinear	74 Ch: impulse invariance
Spec in dB	$\alpha_p = 0.7, \alpha_s = 14$, so $10^{0.07} - 1 = 0.174898$ and $10^{1.4} - 1 = 24.118864$	
Analog edges	pre-warp: $\Omega_p = 2 \tan(0.075\pi) = 0.480158$ $\Omega_s = 2 \tan(0.30\pi) = 2.752764$	linear: $\Omega_p = 0.15\pi = 0.471239$ $\Omega_s = 0.6\pi = 1.884956$
Ω_s/Ω_p	5.733043	4.000000
N exact	1.4106	1.7769
N	2	2
Ω_c	0.742485	0.728694
Poles	$-0.52502 \pm j0.52502$ 0.551284	$-0.51526 \pm j0.51526$ 0.530995
$H_a(s)$	$\frac{s^2 + 1.050033s + 0.551284}{\text{not needed}}$	$\frac{s^2 + 1.030529s + 0.530995}{A_{1,2} = \pm j0.51526}$
Residues	from the substitution	$e^{p_k T} = 0.51979 \mp j0.29435$
z poles	$0.082883(1 + z^{-1})^2$	$0.303336 z^{-1}$
$H(z)$	$\frac{1 - 1.036997z^{-1} + 0.368530z^{-2}}{0.922571 \text{ (spec } \geq 0.922571) \checkmark}$	$\frac{1 - 1.039570z^{-1} + 0.356818z^{-2}}{0.898611 \text{ (spec } \geq 0.922571) \text{ fails}}$
$ H $ at ω_p	0.922571 (spec ≥ 0.922571) $\checkmark$	0.898611 (spec ≥ 0.922571) fails
$ H $ at ω_s	0.072559 (spec ≤ 0.199526) $\checkmark$	0.191753 (spec ≤ 0.199526) barely
$ H $ at DC	1.000000 exactly	0.956147

- **Read the last three rows: that is the whole comparison, in numbers.** The bilinear filter meets the passband exactly and clears the stopband by a factor of three. The impulse-invariant filter of the same order misses the passband edge, scrapes the stopband, and does not even have unity DC gain.
- Every one of those shortfalls is **aliasing**: the analog tail beyond π/T folds back and adds to the response. It is worst where the response is smallest, which is why the stopband suffers most.
- The cure is to **over-design**: raise N or tighten α_s before designing. Nothing can remove the aliasing itself.
- **Note $b_0 = 0$** : the numerator starts at z^{-1} . That is general, not specific: when $H_a(s)$ is strictly proper by two or more degrees, $h[0] = h_a(0) = 0$, so the impulse-invariant filter always has a one-sample delay built in. **It is not a dropped term.**

81 Ch, 80 Ch, 69 Bh 12

$\omega_p = 0.25\pi$ at 0.5 dB; -15 dB at $\omega_s = 0.55\pi$; $f_s = 1$ Hz so $T = 1$.

- $\Omega_p = 0.25\pi = 0.78540$, $\Omega_s = 0.55\pi = 1.72788$ (**no pre-warp**).
- $\alpha_p = 0.5$, $\alpha_s = 15$; $N = 3.5039 \Rightarrow \boxed{N = 4}$, $\Omega_c = 1.02161$.
- $H_a(s) = \frac{\Omega_c^4}{(s^2 + 0.78191s + 1.04369)(s^2 + 1.88770s + 1.04369)}$, constant term $\Omega_c^2 = 1.04369$ in both.

$$H(z) = \frac{0.088750z^{-1} + 0.174840z^{-2} + 0.023521z^{-3}}{1 - 1.513201z^{-1} + 1.180022z^{-2} - 0.449386z^{-3} + 0.069280z^{-4}}$$

- Checks: $|H(e^{j0.25\pi})| = 0.94338$ against a required $10^{-0.5/20} = 0.94406$, so it **misses the passband by 0.0007**; $|H(e^{j0.55\pi})| = 0.11982 \leq 0.17783 \checkmark$; and $|H(e^{j0})| = 1.00139$, **slightly above 1**.
- Both anomalies are aliasing, and both are worth one line in the answer. **A DC gain above 1 is the signature of impulse invariance** and never happens with the bilinear method.

70 Asa 13

$\omega_p = 0.2\pi$ at 0.5 dB; -15 dB at 0.35π ; $T = 1$.

- $\Omega_p = 0.62832$, $\Omega_s = 1.09956$; $N = 4.9367 \Rightarrow \boxed{N = 5}$, $\Omega_c = 0.77542$.
- N odd, so there is a real pole at $-\Omega_c$ and two quadratic sections, $(s + 0.77542)(s^2 + 0.47924s + 0.60128)(s^2 + 1.25466s + 0.60128)$.

$$H(z) = \frac{0.0069166z^{-1} + 0.044472z^{-2} + 0.027021z^{-3} + 0.0015381z^{-4}}{1 - 2.584442z^{-1} + 2.999693z^{-2} - 1.857044z^{-3} + 0.603063z^{-4} - 0.081324z^{-5}}$$

- $|H(e^{j0.2\pi})| = 0.94403$ (need 0.94406), $|H(e^{j0.35\pi})| = 0.17189$ (need ≤ 0.17783), $|H(e^{j0})| = 1.00002$. **The same three aliasing signatures**, smaller because Ω_s/Ω_p is smaller here.

78 Bh, 72 Ch 12+3

200 Hz and 500 Hz, 5 dB and 12 dB, $f_s = 5000$ Hz. **The only impulse-invariance paper where $T \neq 1$** , so the one where the convention matters. 78 Bh's rider is pre-warping, 72 Ch's is the bilinear/impulse-invariance comparison.

- $T = 1/5000$. $\Omega_p = 2\pi(200) = 1256.64$, $\Omega_s = 2\pi(500) = 3141.59$ (rad/s, **not** divided by anything: $\Omega = \omega/T = 2\pi f$).
- $10^{0.5} - 1 = 2.1623$, $10^{1.2} - 1 = 14.849$, **ratio 6.8672**; $N = 1.0514 \Rightarrow \boxed{N = 2}$, $\Omega_c = 1036.29$.
- Poles $-732.769 \pm j732.769$; residues $A_{1,2} = \mp j732.769$; $e^{p_k T} = 0.854421 \mp j0.126123$.

$$H(z) = \frac{184.838 z^{-1}}{1 - 1.708842z^{-1} + 0.745942z^{-2}} \quad (\text{Proakis, } h[n] = h_a(nT))$$

$$H(z) = \frac{0.0369676 z^{-1}}{1 - 1.708842z^{-1} + 0.745942z^{-2}} \quad (\text{Oppenheim, } h[n] = T h_a(nT))$$

- **The first has a DC gain of 4982.** That is $\approx 1/T = 5000$, and it is exactly why the second convention exists. The poles are identical; only the numerator scales.
- With the T scaling: $|H(e^{j0.08\pi})| = 0.5633$ against a required $10^{-5/20} = 0.5623$ ✓ (by a hair), and $|H(e^{j0.2\pi})| = 0.1114$ against $10^{-12/20} = 0.2512$ ✓.
- With $T = 1$ the two conventions coincide, which is why the other eight impulse-invariance papers never raise the question. **Here you must say which you used.**

67 Mng 15

QUESTION AS PRINTED 2067 Mangsir · [15]

Find the system function for a digital filter using the impulse invariance technique from the analog Butterworth filter transfer function $H(s) = \frac{1}{(s+1.3)(s-1.3e^{j2\pi/3})(s-1.3e^{-j2\pi/3})}$ with $T = 1$ second, and draw the block diagram of the system function $H(z)$ realized in terms of second order sections.

- **Recognise it before computing.** $1.3e^{\pm j2\pi/3} = -0.65 \pm j1.12583$, so the three poles sit at $120^\circ, 180^\circ, 240^\circ$ on a circle of radius 1.3. **This is a third-order Butterworth filter with $\Omega_c = 1.3$** , already designed. Steps 1 to 4 are done for you.
- Expanded: $H_a(s) = \frac{1}{s^3 + 2.6s^2 + 3.38s + 2.197}$, and $2.197 = 1.3^3$, which confirms the reading.
- Residues $A_k = 1/\prod_{j \neq k} (p_k - p_j)$:

Pole p_k	Residue A_k	z pole $e^{p_k T}, T = 1$
-1.30000	+0.591716	0.272532
-0.65000 + $j1.12583$	-0.295858 + $j0.170814$	0.224701 - $j0.471212$
-0.65000 - $j1.12583$	-0.295858 - $j0.170814$	0.224701 + $j0.471212$

- The residues sum to zero, which is the general reason $b_0 = 0$ below.

$$\begin{aligned} H(z) &= \frac{0.591716}{1 - 0.272532z^{-1}} + \frac{-0.591716 + 0.293938z^{-1}}{1 - 0.449403z^{-1} + 0.272532z^{-2}} \\ &= \frac{0.189281z^{-1} + 0.081154z^{-2}}{1 - 0.721935z^{-1} + 0.395008z^{-2} - 0.074274z^{-3}} \end{aligned}$$

- **The block diagram the paper asks for is the parallel form**, and it falls out of the partial fraction expansion with no extra work: one first-order section and one second-order section, both fed by $x[n]$, their outputs summed.
- **Draw it as two direct-form-II boxes in parallel:** the real section has one delay and one feedback multiplier 0.272532; the quadratic section has two delays, feedback multipliers 0.449403 and -0.272532 , and feedforward multipliers -0.591716 and 0.293938.
- Why parallel and not cascade: impulse invariance *produces* a sum of terms. Forcing it into a cascade means factorising the numerator for no reason. Cross-reference chapter 4 for the structures themselves.

3 Bilinear transformation, Chebyshev prototype

HOT 4/45 72 Ma **11**, 74 Ma **9**, 70 Ma **9**, 75 Bh **10** · **39 marks, all four on the EX 753 papers**

Steps 1, 2, 6 and 7 are identical to §1. **Only steps 3 to 5 change:** a different order formula, $\Omega_c = \Omega_p$ with no formula, and poles on an ellipse instead of a circle.

QUESTION AS PRINTED 2072 Magh · [11]

Design a low pass discrete time Chebyshev filter by applying bilinear transformation for $0.707 \leq |H(e^{jw})| \leq 1$ for $|w| \leq 0.2\pi$ and $|H(e^{jw})| \leq 0.1$ for $0.5\pi \leq |w| \leq \pi$.

Step 1 to 3: spec, pre-warp, ripple factor

- $\alpha_p = -20 \log_{10} 0.707 = 3.0116$ dB, $\alpha_s = -20 \log_{10} 0.1 = 20$ dB.
- $\Omega_p = 2 \tan(0.1\pi) = \mathbf{0.64984}$, $\Omega_s = 2 \tan(0.25\pi) = \mathbf{2}$ exactly.

$$\varepsilon = \sqrt{10^{0.1\alpha_p} - 1} = \sqrt{10^{0.30116} - 1} = \mathbf{1.00030} \approx 1$$

- $\varepsilon = 1$ **because 0.707 is the 3 dB point**, the same coincidence 73 Ch exploits in §1.3.
- $\Omega_c = \Omega_p = 0.64984$. A Chebyshev filter ripples right up to the passband edge, so there is no separate cut-off to compute.

Step 4: order, by the $\cosh^{-1}$ formula

$$N \geq \frac{\cosh^{-1} \sqrt{(10^2 - 1)/(10^{0.30116} - 1)}}{\cosh^{-1}(2/0.64984)} = \frac{\cosh^{-1}(9.94687)}{\cosh^{-1}(3.07769)} = 1.6694$$

$$\boxed{N = 2}$$

- Use $\cosh^{-1} x = \ln(x + \sqrt{x^2 - 1})$ if the calculator lacks the key: $\ln(19.8433) = 2.98787$ and $\ln(5.9884) = 1.78982$.
- **Compare: a Butterworth filter on this same spec needs $N = 3$.** That one saved order is the whole argument for Chebyshev.

Step 5: the pole ellipse

$$\alpha = \frac{1}{\varepsilon} + \sqrt{1 + \frac{1}{\varepsilon^2}} = 1.0003 + 1.4145 = \mathbf{2.41370}$$

$$a = \frac{1}{2}(\alpha^{1/2} - \alpha^{-1/2}) = \mathbf{0.45497}, \quad b = \frac{1}{2}(\alpha^{1/2} + \alpha^{-1/2}) = \mathbf{1.09864}$$

- Semi-axes: $a\Omega_c = 0.29566$ (real direction), $b\Omega_c = 0.71394$ (imaginary direction). **Taller than wide, always.**
- $N = 2$, so $\theta_k = \frac{(2k+1)\pi}{4} = 45^\circ, 135^\circ$:

$$s_k = -a\Omega_c \sin \theta_k + j b\Omega_c \cos \theta_k = \mathbf{-0.20906 \pm j0.50483}$$

$$H_a(s) = \frac{K}{s^2 + 0.418125s + 0.298560}, \quad K = \frac{|s_k|^2}{\sqrt{1 + \varepsilon^2}} = \frac{0.298560}{1.41443} = \mathbf{0.211082}$$

- **The $\sqrt{1 + \varepsilon^2}$ divisor is because N is even.** Omit it and the passband sits a factor of 1.414 too high everywhere.

Step 6 and 7: map back and check

$$\boxed{H(z) = \frac{0.041108 (1 + z^{-1})^2}{1 - 1.441705z^{-1} + 0.674282z^{-2}}}$$

- $|H(e^{j0.2\pi})| = \mathbf{0.70700}$, exactly the specified 0.707. ✓
- $|H(e^{j0.5\pi})| = \mathbf{0.05563} \leq 0.1$. ✓
- $|H(e^{j0})| = \mathbf{0.70700}$, **not 1**. The passband oscillates between 0.707 and 1 and, with N even, it **starts at the bottom of the ripple**. This is correct and looks wrong: say so in the answer.

The other three Chebyshev papers

Paper	Spec	ϵ	N	Poles	$H(z)$
74 Ma	0.2π , 1 dB; -15 dB at 0.3π ; $T=1$	0.50885	4	$-0.0907 \pm j0.6390$ $-0.2189 \pm j0.2647$	$K = 0.0018356$, numerator $K(1 + z^{-1})^4$, denominator $1 - 3.054340z^{-1}$ $+ 3.828999z^{-2}$ $- 2.292452z^{-3}$ $+ 0.550745z^{-4}$
70 Ma	0.25π , 0.55π ; $\alpha_{\max}=1.01$, $\alpha_{\min}=13.55$; $f_s=0.5$ Hz	0.51169	2	$-0.45286 \pm j0.74042$	$K = 0.102154$, numerator $K(1 + z^{-1})^2$, denominator $1 - 0.989132z^{-1}$ $+ 0.448133z^{-2}$

- 74 Ma is the §1.2 spec again ($0.2\pi/0.3\pi$, 1 dB/15 dB) done with Chebyshev: $N = 4$ where Butterworth needed $N = 6$. **That is the comparison 76 Bh asks for, with real numbers.**
- 70 Ma prints $\alpha_{\max}$ for the passband and $\alpha_{\min}$ for the stopband, which is the reverse of the usual naming. It means $\alpha_p = 1.01$, $\alpha_s = 13.55$: the $\delta_p = 0.11$, $\delta_s = 0.21$ spec yet again, in a third notation.
- Both have even N , so both have DC gain $1/\sqrt{1 + \epsilon^2}$, not 1.

75 Bh 10

an analog design first

QUESTION AS PRINTED 2075 Bhadra · [10]

Design an *analog* filter using Chebyshev approximation with maximum passband attenuation 5 dB at 20 rad/sec and stopband attenuation 20 dB at 50 rad/sec; then, taking a sampling frequency of 0.5 Hz, convert the obtained analog filter to digital using the bilinear transformation method.

- **The edges are already analog**, in rad/s. **Do not pre-warp**: the question specifies the analog filter directly, so pre-warping would design the wrong one. This is the only paper in the chapter where the pre-warp is skipped legitimately.
- $\Omega_p = 20$, $\Omega_s = 50$, $\alpha_p = 5$, $\alpha_s = 20$, $\epsilon = \sqrt{10^{0.5} - 1} = \mathbf{1.47047}$.

$$N \geq \frac{\cosh^{-1}(6.76647)}{\cosh^{-1}(2.5)} = 1.6592 \Rightarrow \boxed{N = 2}$$

- $\alpha = 1.88938$, $a = 0.32352$, $b = 1.05103$; semi-axes 6.47037 and 21.02060; poles $-\mathbf{4.57524} \pm \mathbf{j14.86381}$.

$$H_a(s) = \frac{136.011}{s^2 + 9.15049s + 241.866}$$

- Now $T = 2$ s, and **here T does NOT cancel**, because the analog filter was fixed before the mapping instead of being derived through a pre-warp. Substitute $s = \frac{2}{2} \frac{1-z^{-1}}{1+z^{-1}} = \frac{1-z^{-1}}{1+z^{-1}}$:

$$H(z) = \frac{0.539692(1 + z^{-1})^2}{1 + 1.911510z^{-1} + 0.927382z^{-2}}$$

- **Both denominator coefficients are positive**, which in §1 never happened. The poles are at $-0.956 \pm j0.110$, near $z = -1$: sampling a 20 rad/s filter at 0.5 Hz is sampling far below Nyquist, and the filter that results is a high-frequency one. **Say that the sampling rate is inconsistent with the analog design** rather than presenting the answer as a good low-pass filter.

4 Spectral transformation

HOT 6/45 70 Ch **5**, 80 Ba **4**, 70 Asa **2**, 79 Bh, 71 Ch, 69 Ch **4** · the 79 Bh family's 4 marks are worked at the end of §1

QUESTION AS PRINTED 2070 Chaitra · [5]

A digital LPF with cut off frequency $w_c = 0.2575\pi$ is given as $H(z) = \frac{0.1 + 0.4z^{-1}}{1 - 0.6z^{-1} + 0.1z^{-2}}$. Design a digital high pass filter with $w'_c = 0.3567\pi$.

- **Sanity check the data first**, because it is free: $|H(e^{j0.2575\pi})| = 0.70789 \approx 1/\sqrt{2}$, so 0.2575π really is this filter's 3 dB point. **The paper's numbers are consistent.**

$$\alpha = -\frac{\cos \frac{\omega_c + \omega'_c}{2}}{\cos \frac{\omega_c - \omega'_c}{2}} = -\frac{\cos(0.3071\pi)}{\cos(0.0496\pi)} = -\frac{0.56928}{0.98788} = -\mathbf{0.57658}$$

- $|\alpha| < 1 \checkmark$, so the map is stable.
- Substitute $z^{-1} \rightarrow -\frac{z^{-1} + \alpha}{1 + \alpha z^{-1}} = -\frac{z^{-1} - 0.57658}{1 - 0.57658z^{-1}}$, multiplying top and bottom by $(1 - 0.57658z^{-1})^2$ to clear the fractions:

$$H_{hp}(z) = \frac{0.48106 - 0.94325z^{-1} + 0.38393z^{-2}}{1 - 0.68240z^{-1} + 0.12585z^{-2}}$$

- Check: $|H_{hp}(e^{j0.3567\pi})| = \mathbf{0.70789}$, **the same value the low pass had at its own cut-off**. That is exactly what the transformation promises, and it is the one line that proves the answer.
- **Note the numerator's signs alternate** where the prototype's did not. A high pass built this way pushes the zeros towards $z = +1$.

80 Ba, 70 Asa **4** **2**

Both are bookwork, answered in §6.6 of the theory: why the transformation is required (impulse invariance cannot make a high pass; doing it in the z domain avoids aliasing entirely), the all-pass form the map must take, and the LP $\rightarrow$ HP row of the table with its α .

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full. $\rightarrow$ reads “leads to / therefore”, $+$ reads “and”, $=$ reads “is”. $\checkmark$ means the property holds, $\times$ that it does not.

Short	Means	Short	Means
DSP	digital signal processing	DSAP	digital signal analysis and processing
DT	discrete-time	CT	continuous-time
LTI	linear time-invariant	LSI	linear shift-invariant, the same thing
TI	time-invariant	BIBO	bounded input, bounded output
LCCDE	linear constant-coefficient difference equation	ROC	region of convergence
DTFS	discrete-time Fourier series	DTFT	discrete-time Fourier transform
CTFT	continuous-time Fourier transform	DFS	discrete Fourier series
DFT	discrete Fourier transform	IDFT	inverse DFT
FFT	fast Fourier transform	DIT	decimation in time
DIF	decimation in frequency	MAC	multiply and accumulate
FIR	finite impulse response	IIR	infinite impulse response
LPF	low pass filter	HPF	high pass filter
BPF	band pass filter	BSF	band stop filter
BLT	bilinear transformation	II	impulse invariance
A/D	analog to digital	D/A	digital to analog
S/H	sample and hold	ZOH	zero-order hold
SNR	signal to noise ratio	PYQ	past-year question
coeff	coefficient	freq	frequency
eqn	equation	no.	number
ops	operations	attr	attribute
w/	with	w/o	without
Eg	for example	Vs	compare / differentiate
+Fig	a figure is wanted	(I)	importance / need
(+)	advantage	(-)	disadvantage

Exam paper codes

Month codes

- **Ba** Baishakh **Jth** Jestha **Asa** Ashad
- **Shr** Shrawan **Bh** Bhadra **Ash** Ashwin
- **Ka** Kartik **Mng** Mangsir **Po** Poush
- **Ma** Magh **Ch** Chaitra
- **Ash and Asa are different months:** 2075 Ashwin and 2070 Ashad are separate papers.

How a year code is written

- **Bold** = **Regular** exam. Plain = Back exam.
 - “New Back (2066 & Later Batch)” counts as Back.
 - A header printing “Regular / Back” in one box counts as Regular.
- Typewriter = the EX 753 paper (DSP, BEX, IV/II).
Normal font = **CT 704** (DSAP, BEI/BCT, IV/I).
- The two axes are independent: **76 Ch** is CT 704 Regular, 76 Ash is CT 704 Back, 76 Bh is EX 753 Regular, 74 Ma is EX 753 Back.
- **69 Bh and 69 Bh are two different papers**, one in each course.

Frequency chips

- **TOP n/45** asked in **10 or more** of the 45 papers.
- **HOT n/45** asked in **4 to 9** papers.
- **PIN n/45** asked in **1 to 3** papers.
- **verified/added** written or corrected against a reference because the source notes did not cover it, or covered it wrongly.
- **5** marks the question has actually carried. Several chips mean the same question has been set at different marks in different years.
- The count is over **45 papers**: 28 CT 704 (2066 Magh to 2082 Bhadra) and 17 EX 753 (2069 Bhadra to 2081 Chaitra).
- Counts come from the Detailed PYQ document, which reproduces every question in full.